

CSE475 Machine Learning

Task 1 Report: Data Understanding and Related Work

GNN-Based EMG Gesture Recognition on NinaPro DB1

Group 04 | Track 1 — Graph Neural Network | East West University

1. Dataset Overview

This project uses NinaPro DB1, a public surface EMG dataset recorded from 10 forearm electrodes while subjects performed a series of hand and finger gestures. Each timepoint is labelled with the gesture being performed (stimulus/restimulus), the repetition number, and the subject ID.

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Property	Value
Dataset	NinaPro DB1
Source	ninapro.hevs.ch (Kaggle:mansibmursalin/ninapro-db1-full-dataset)
Application area	EMG gesture recognition
EMG channels	10
Missing values	0
Duplicate rows	0
Track	Track 1 — GNN on table-style data

2. Theoretical Background: Graph Neural Networks for EMG Gesture Recognition

2.1 Why Graph Neural Networks?

Surface electromyography (sEMG) signals recorded from multiple electrodes naturally exhibit spatial relationships—muscle activity from a single gesture propagates across adjacent and functionally connected channels. Traditional machine learning approaches treat each channel independently or concatenate features into flat vectors, discarding this spatial structure entirely. This is where Graph Neural Networks (GNNs) offer a fundamental advantage.

In the context of EMG gesture recognition, this means:

- Channels that exhibit correlated activity during a particular gesture can share information
- The model can learn which channel relationships are most informative for discriminating between gestures
- Spatial patterns unique to each gesture are captured automatically

2.2 Graph Construction Strategies

The choice of graph topology—how edges between channels are defined—is critical. Three primary strategies exist in the literature:

1. Fixed Physical Adjacency: Edges connect channels that are physically adjacent on the electrode grid. This is common in HD-sEMG setups where channels form a dense array but becomes questionable in sparse configurations where electrodes are far apart.
2. Correlation-Driven Topology: Edges are defined based on signal similarity, typically using Pearson correlation coefficients between channel activity over time. This captures functional connectivity—channels that activate together during gestures—which is more relevant than mere physical proximity.
3. Learned Topology: The graph structure itself becomes a trainable parameter, allowing the model to discover optimal connections during training.

Our work adopts the correlation-driven approach** as it offers the best balance between exploiting spatial information and adapting to the sparse 10-electrode configuration of NinaPro DB1.

2.3 Addressing Class Imbalance

A defining characteristic of raw EMG streams is the overwhelming dominance of the "rest" class—subjects spend far more time at rest than actively performing gestures. In NinaPro DB1, the rest class outnumbers any single gesture class by 30-80×. This severe class imbalance presents challenges:

- Standard accuracy metrics become misleading (a model predicting "rest" for all windows would achieve high accuracy)
- Gradient updates are dominated by the majority class, leading to poor gesture recognition performance
- Standard cross-entropy loss struggles with minority classes

Our approach addresses this through:

- Class-weighted loss functions that penalize misclassification of minority gestures more heavily
- Macro-F1 evaluation rather than raw accuracy
- Per-class recall analysis to ensure all gestures are represented

2.4 Subject-Independent Evaluation

To ensure generalizability, we employ a subject-independent (leave-one-subject-out) evaluation protocol. This prevents data leakage where training and test sets contain data from the same subject, which would artificially inflate performance. For a dataset with N subjects:

- Train: N-1 subjects' data
- Test: The held-out subject's data
- This process repeats N times, with the average performance reported

3. Feature Extraction

Raw EMG timepoints carry little individual meaning, since a gesture is expressed as a pattern over time rather than a single instant. The continuous signal was segmented into overlapping windows (200 samples, 100-sample step), and six standard time-domain descriptors were computed per channel per window:

Feature	Description
RMS	Root Mean Square - measure of signal amplitude
MAV	Mean Absolute Value - average absolute amplitude
WL	Waveform Length - cumulative variation over the window

VAR	Variance - spread of signal values
ZC	Zero Crossings - number of times the signal crosses zero
SSC	Slope Sign Changes - number of sign changes in signal slope

This produced the tabular feature set used for the EDA and subsequent GNN modeling. For a 10-channel setup with 6 features per channel, each window is represented as a feature vector of length 60.

4. Exploratory Data Analysis - Key Findings

Class balance: The "rest" class dominates, outnumbering any single gesture class by roughly 30-80×, motivating macro-F1 and per-class recall over raw accuracy.

Data quality: No missing values or duplicates; ZC is near-constant (uninformative) on channels 2, 4, and 7.

Redundancy: MAV is highly correlated (>0.95) with RMS on nearly every channel, and WL overlaps with both on several channels—candidates for removal before modeling.

Distributions: RMS/MAV/VAR features are strongly right-skewed, reflecting mostly low-activity windows with occasional high-amplitude contraction bursts.

Separability: PCA/t-SNE show partial, non-linear class separation, suggesting a linear classifier alone is unlikely to be sufficient—supporting the case for a GNN.

Subject structure: Window counts vary by subject and class, confirming a subject-wise (grouped) train/test split is required to avoid data leakage.

5. Related Work

Ten studies (2022–2026) on GNN/GCN-based EMG and physiological-signal gesture recognition were reviewed. A summary of these works, their contributions, limitations, and their relationship to our research is presented in the table below.

Table 1: Summary of related work (2022–2026) on GNN/GCN-based physiological-signal gesture recognition.

Authors (Year)	Title	Dataset	Application Area	Method/Model	Metrics	Strengths	Limitations / Gap	Relation to Our Work
Xu, Chen, Ruan, & Zhang [1] (2023)	"Research on electromyography pattern recognition based on graph neural network"	HD-sEMG (multi-subject)	Cross-user EMG pattern / gesture recognition	Spatio-temporal GCN (CNN-MSTGCN)	Accuracy, F1-score	Combines CNN feature extraction with multi-scale spatio-temporal graph module; targets cross-user generalization	Channel-graph topology is fixed rather than learned per gesture; HD-sEMG focus limits applicability to sparse electrode setups	Our work uses correlation-driven graph topology on low-channel NinaPro DB1, enabling adaptive graph learning rather than fixed physical adjacency
Lee et al. [2] (2023)	"Stretchable array electromyography sensor with graph neural network for static and dynamic gestures recognition system"	Custom stretchable 18-gesture array	Static/dynamic hand gesture recognition	GNN on flexible-array electrodes	~97% accuracy	Pairs stretchable sensor array with spatially-aware graph model; high accuracy on dynamic gestures	Custom hardware/dataset not compatible with public EMG benchmarks; limited reproducibility	We use public NinaPro DB1 benchmark to ensure reproducibility and comparability with existing methods
Massa, Riboni, & Nazarpour [3] (2023)	"Explainable AI-powered graph neural networks for HD EMG-based gesture intention recognition"	HD-sEMG, 65 gestures, 20 subjects	Fine-grained gesture-intention recognition	GNN + GNNExplainer topology refinement (xAI-GNN)	~8.75% average error	Uses explainable AI to prune uninformative edges, improving interpretability and efficiency	Preliminary version reported high variance across gestures; HD-sEMG only; no class imbalance handling	Our approach addresses class imbalance explicitly in training strategy, applicable to standard 10-electrode configuration
Lee, Jiang, Yang, Yang, & Zhao [4] (2025)	"Decoding gestures in electromyography: Spatiotemporal graph neural networks for generalizable and interpretable classification"	Five public EMG datasets	Generalizable, interpretable gesture classification	Spatiotemporal GCN	State-of-the-art accuracy across datasets	Graph encodes both spatial electrode proximity and temporal adjacency; strong cross-dataset performance	Prioritizes cross-dataset generalization over single-subject, low-channel deployment; fixed graph topology	Our work focuses specifically on single-subject, low-channel NinaPro DB1 with correlation-driven graph construction
Massa, Riboni, & Nazarpour [5] (2022)	"Graph neural networks for HD EMG-based movement intention recognition: An initial investigation"	HD-sEMG, 65 gestures, 20 subjects	Movement-intention recognition (initial study)	Baseline GNN on HD-EMG channel graph	Preliminary accuracy metrics	Early proof-of-concept that GNNs are viable for raw-EMG-channel gesture decoding	Simple, non-adaptive graph topology; only two HD-EMG arrays tested	We extend from HD-sEMG to low-channel sEMG and use correlation-based adaptive graph topology

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Vipulanandan, Premaratne, & Murthi [6] (2024)	"A graph neural network model for real-time gesture recognition based on sEMG signals"	Myoband sEMG, 8 electrodes, 8 subjects	Real-time hand gesture recognition	GNN on correlation-based signal graph	99% accuracy	Graph built directly from inter-channel signal correlation, not fixed layout	Evaluated on only 5 gesture classes and a small subject pool; limited generalization assessment	We extend correlation-based graph construction to 10 channels and 12 gesture classes with rigorous subject-independent evaluation
Li, Ji, Yu, Li, Jin, Liu, Bai, & Ye [7] (2023)	"A sequential learning model with GNN for EEG-EMG-based stroke rehabilitation BCI"	EEG + EMG, stroke rehabilitation cohort	Multimodal stroke rehabilitation BCI	GIN (Graph Isomorphism Network) on heterogeneous graph	Classification accuracy, AUC	Combines EEG and EMG as heterogeneous graph using mutual information for edges; strong gains over baselines	Heterogeneous EEG-EMG graph adds complexity not needed for EMG-only setups; stroke-specific context	We focus on EMG-only graph construction for gesture recognition, avoiding unnecessary multimodal complexity
Quadrelli, Canepa, Di Domenico, Boccardo, Chiappalone, & Laffranchi [8] (2025)	"Advances in HD-EMG interfaces and spatial algorithms for upper limb prosthetic control"	HD-EMG interfaces (review scope)	Upper-limb myoelectric prosthetic control	Review of GNN and spatial-ML approaches	N/A (it is a review paper)	Surveys spatial/graph-based ML exploiting HD-EMG electrode layout to improve intention detection	Mini-review scope; does not benchmark a single model on standard low-channel dataset	Provides context for graph-based EMG research but highlights gap in low-channel sEMG GNN applications
Klepl, He, Wu, Blackburn, & Sarrianni [9] (2024)	"Graph neural network-based EEG classification: A survey"	Cross-dataset (EEG classification survey)	Systematic review of GNN-based physiological classification	Survey of spectral and spatial GNN variants	N/A (it is a review paper)	Identifies that spectral graph convolutions dominate current GNN-based biosignal classifiers	Focused on EEG; does not cover low-channel sEMG graph construction specifically	Emphasizes the need for EMG-specific GNN studies; our work addresses this gap for sEMG
Hou, Jia, Lun, Hao, Shi, Li, Zeng, & Lv [10] (2022)	"GCNs-Net: A graph convolutional neural network approach for decoding time-resolved EEG motor imagery signals"	EEG motor-imagery benchmark datasets	Time-resolved motor-imagery decoding	GCNs-Net (graph convolutions + pooling)	Accuracy, Kappa score	Learns generalized features respecting electrode topology; robust across individual variability	Designed for EEG electrode topology; not validated on sparse EMG channel graphs	We adapt graph convolution principles from EEG to EMG, with topology informed by signal correlation rather than physical layout

6. Research Gap

Identified Gaps in Current Literature

1. HD-sEMG Concentration: Existing GNN/GCN work on EMG gesture recognition is concentrated on high-density (HD) sEMG setups with dozens

to over a hundred electrodes [1, 3, 4]; standard low-channel (10-electrode) surface EMG benchmarks such as NinaPro DB1 remain largely unexamined with GNNs.

2. Fixed vs. Learned Topology: Graph topology in most prior work is fixed by physical electrode adjacency rather than learned from inter-channel signal correlation, which is the more relevant cue when only 10 sparsely-placed channels are available [5].
3. Hardware-Specific Studies: Custom-hardware studies [2] report strong accuracy but on non-public, non-comparable datasets, limiting reproducibility against standard EMG benchmarks.
4. EEG-Dominated Research: Most GNN-based bio-signal survey and review work to date centers on EEG [9, 10], with comparatively few systematic studies applying the same graph-construction principles to low-channel sEMG.
5. Class Imbalance Neglect: None of the reviewed studies explicitly address severe rest-vs-gesture class imbalance within the graph construction or training strategy itself, despite this being a dominant characteristic of raw EMG streams.
6. Limited Generalization Assessment: Most studies evaluate on small gesture sets (≤ 5 classes) or limited subject pools [6], lacking rigorous subject-independent evaluation on diverse gesture repertoires.

7. Our Contribution

This work closes these gaps by:

1. Building a correlation-driven channel graph from a standard low-density NinaPro configuration
2. Evaluating under a subject-independent, leakage-free split with explicit class-imbalance handling
3. Providing a reproducible benchmark on public NinaPro DB1 data
4. Extending graph-based EMG research from HD-sEMG to low-channel sEMG applications

5. Addressing the rest-vs-gesture class imbalance through weighted loss functions and macro-F1 evaluation
6. Demonstrating generalization across all 12 gesture classes with rigorous subject-independent validation

Comparison Framework

Our work is positioned relative to existing literature along the following dimensions:

Dimension	Existing Work	Our Work
Electrode Configuration	HD-sEMG (32-128+ channels)	Low-channel (10 electrodes)
Graph Construction	Fixed physical adjacency	Correlation-driven topology
Dataset Type	Custom/hardware-specific	Public (NinaPro DB1)
Evaluation Protocol	Subject-dependent or limited	Subject-independent (LOSO)
Class Imbalance	Not addressed	Explicit weighting strategy
Gesture Classes	5-18 classes	12 classes + rest
Reproducibility	Limited (custom hardware)	High (public dataset)

8. Summary

The EDA confirms that NinaPro DB1 requires windowed feature extraction, contains redundant amplitude-related features, and exhibits severe class imbalance dominated by the rest class. Combined with the identified research gap—the lack of GNN-based approaches for low-channel sEMG with correlation-driven graph construction and explicit imbalance handling—these findings directly inform our proposed methodology. Our work will implement a correlation-based channel graph with a GCN architecture, evaluated under a subject-independent protocol, providing a reproducible benchmark for future research in this domain.